Sahaj Bhadja

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Summary

Engineer with 2+ years building automation, analytics, and AI-assisted workflows across APIs, data pipelines, and SaaS products using SQL, Python, experimentation, and instrumentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built ETL-style integration pipelines across FastAPI, SQL Server, Supabase, and cloud storage, processing 5K+ monthly jobs with validation checks, reconciliation logic, and CI/CD-backed releases for reliable operations.
- Designed and optimized a multi-tenant PostgreSQL schema for 200+ users and 10K+ records, using indexing, query tuning, and data validation workflows that reduced latency by 35% in production.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented a production backend on AWS serving 150+ instructors and 10K+ grading requests daily, with CI/CD workflows, instrumentation, and stable API performance during high-usage operational cycles.
- Developed React interfaces and backend flows for uploads, grading history, rubric generation, and concurrent evaluation, reducing feedback turnaround by 60% while improving visibility for content-heavy users.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Troubleshot API integrations with AWS Glue and S3, improving response times by 30% while leading ETL workflow changes that boosted pipeline efficiency 68% and reduced technical debt 20%.
- Maintained 20+ automated tests and CI pipelines with Jenkins, PyTest, and Mockito, increasing release reliability by 68% and reducing manual work across recurring validation and deployment flows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a resource management platform with REST APIs, WebSockets, and microservices, supporting six workflows and 2M+ synchronized records while reducing code-review turnaround by 25% across release cycles.
- Used Azure SQL, Jenkins, and analytics instrumentation to reduce resource wastage by 42%, increase lead conversions by 15%, and improve release readiness through repeatable workflows.

LST Capitals **Los Angeles, CA**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for booking management, pricing, coupon validation, payments, and status updates across 50K+ booking records, maintaining sub-200ms responses across mobile and web workflows.
- Integrated analytics-aware payment and booking flows with webhook automation and Firebase sync, reducing manual handling by 70% while improving observability across customer-facing touchpoints.

Skills

GTM & Analytics: SQL, pipeline analysis, conversion analysis, segmentation, experimentation, campaign performance, funnel reporting, messaging iteration
Automation & Platforms: Python, workflow automation, API integrations, ETL pipelines, CI/CD, Jenkins, GitHub Actions, Salesforce concepts, HubSpot concepts, Apollo concepts, Common Room concepts
Technical Stack: JavaScript, TypeScript, FastAPI, Flask, REST APIs, PostgreSQL, SQL Server, Azure SQL, AWS, S3, Docker, Git, Agile/Scrum

Projects

PiXMod Live | *Next.js, Stripe, MongoDB*
- Built subscription billing, scalable image storage, and a responsive SaaS interface, increasing premium conversion by 18% while strengthening monetization and self-serve product workflows.

FinNexus Live | *Next.js, TypeScript, Kubernetes*
- Built a fintech platform supporting 50,000+ users and 1M+ transactions, combining real-time updates and secure workflows to support analytics-ready operational visibility at scale.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL*
- Engineered a cloud ingestion and indexing pipeline processing 50K+ records, implementing incremental updates, metadata persistence, and tuning to improve retrieval speed for AI-assisted workflows.